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Bucket Generation

Use your bucket-based bigram model to generate new text by picking tokens at random.

You will need

- your completed bucket model from *Bucket Training*
- pen & paper for writing down the generated “output text”

Your goal

To generate new text from your bucket language model. **Stretch goal:** keep going, generating as much text as possible. Write a whole book!

Key idea

Language models generate text by predicting one word at a time based on learned patterns. Each bucket contains all the tokens that could come next — and some tokens appear multiple times, making them more likely to be picked. Choosing randomly from a bucket and repeating word by word creates new text.



Algorithm

1. **choose a starting bucket** — pick any bucket and write down its label as the first word
2. **close your eyes and pick a random token** from inside that bucket
3. **write down the token** you picked
4. **put the token back** in the bucket (so you can use it again later)
5. **find the bucket** whose label matches the token you just picked
6. **repeat** from step 2 until you reach the desired length or a natural stopping point (e.g. an empty bucket)

Example

Using the same bucket model from the example in *Bucket Training*:

Bucket	Tokens inside
see	spot spot
spot	run jump
run	.
.	see
jump	.

- choose **see** as your starting bucket; write down “see”
- pick from “see” bucket: both tokens are **spot**, so we get **spot**; write it down
- pick from “spot” bucket: randomly between **run** and **jump**
- let’s say we pick **run**; write it down
- pick from “run” bucket: only **.** inside; write it down
- pick from “.” bucket: only **see** inside; write it down
- pick from “see” bucket: get **spot**; write it down
- pick from “spot” bucket: this time we pick **jump**; write it down
- pick from “jump” bucket: only **.** inside; write it down

After the above steps, the generated text is “see spot run. see spot jump.”

The randomness comes from physically picking tokens without looking. Buckets with more tokens of the same type are more likely to produce that token.